Numerical Method - Homework 1

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(a)

(b)

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Theorem 2.1 (Isomorphism extension theorem)**.** *Given any field F, an algebraic extension field E of F and an isomorphism φ mapping F onto a field F' then φ can be extended to an isomorphism τ mapping E onto an algebraic extension E' of F' (a subfield of the algebraic closure of F').*

Theorem 2.2. *Let F be a field of characteristic 0 and let $f(x) \in F[x]$, and let G be the Galois group of $f(x)$ over F. If $f(x)$ is irreducible, G acts transitively on the roots of $f(x)$.*

Theorem 2.3. *Symmetric Group is Generated by Transposition and n-Cycle.*